

EE 316

Section 3

Lab 05

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Basic Filters and Frequency Response

Introduction:

The goal of this laboratory is to study the frequency characteristics of low-pass and high-pass filters. The effects of input frequency on signal amplitude and phase angle are examined through theoretical calculations and experimental measurements. The theoretical gain and phase angle are calculated for a range of frequencies, and the cutoff frequency is determined and used to identify the point where the filter gain is approximately -3 dB. A physical circuit is also constructed and tested using laboratory equipment to compare the experimental measurements with the theoretical and simulated results.

Theoretical Analysis:

Equations 1 and 2, given in the lab manual, were used to find the gain and phase angle for a low pass filter with $V_{in} = 4V$, $R = 1k\Omega$, and $C = 1\mu F$. Equation 3 was used to find the cutoff frequency.

$$\left| \frac{V_{out}}{V_{in}} \right| = \frac{1}{\sqrt{1 + (2\pi fRC)^2}} \Rightarrow G = 20 \log \left(\frac{1}{\sqrt{1 + (2\pi fRC)^2}} \right) dB \quad (1)$$

$$\theta = \tan^{-1}(-2\pi fRC) \quad (2)$$

$$f_c = \frac{1}{2\pi RC} \quad (3)$$

Equations 4 and 5 show equations 1 and 2 evaluated at $f = 25Hz$. These calculations were then repeated for varying values of f and used to complete Table 1. Equation 6 shows the cutoff frequency calculation. Figures 1 and 2 show the plots of gain and phase angle versus frequency.

$$G = 20 \log \left(\frac{1}{\sqrt{1 + (2\pi * 25 * 1000 * 10^{-6})^2}} \right) dB = -0.1059 dB \quad (4)$$

$$\theta = \tan^{-1}(-2\pi * 25 * 1000 * 10^{-6}) = -8.9271^\circ \quad (5)$$

$$f_c = \frac{1}{2\pi * 1000 * 10^{-6}} = 159.1594 Hz \quad (6)$$

f (Hz)	Theoretical	
	Gain (dB)	Phase Angles (degrees)
25	-0.1059	-8.9271
50	-0.4088	-17.4406
75	-0.8709	-25.2316
100	-1.4451	-32.1419
150	-2.7606	-43.3038
200	-4.1147	-51.4881
300	-6.5830	-62.0533
500	-10.3621	-72.3432
600	-11.8219	-75.1439
700	-13.0845	-77.1908
800	-14.1940	-78.7483
900	-15.1822	-79.9716
1000	-16.0722	-80.9570

Table #1: Low Pass Filter Theoretical Results

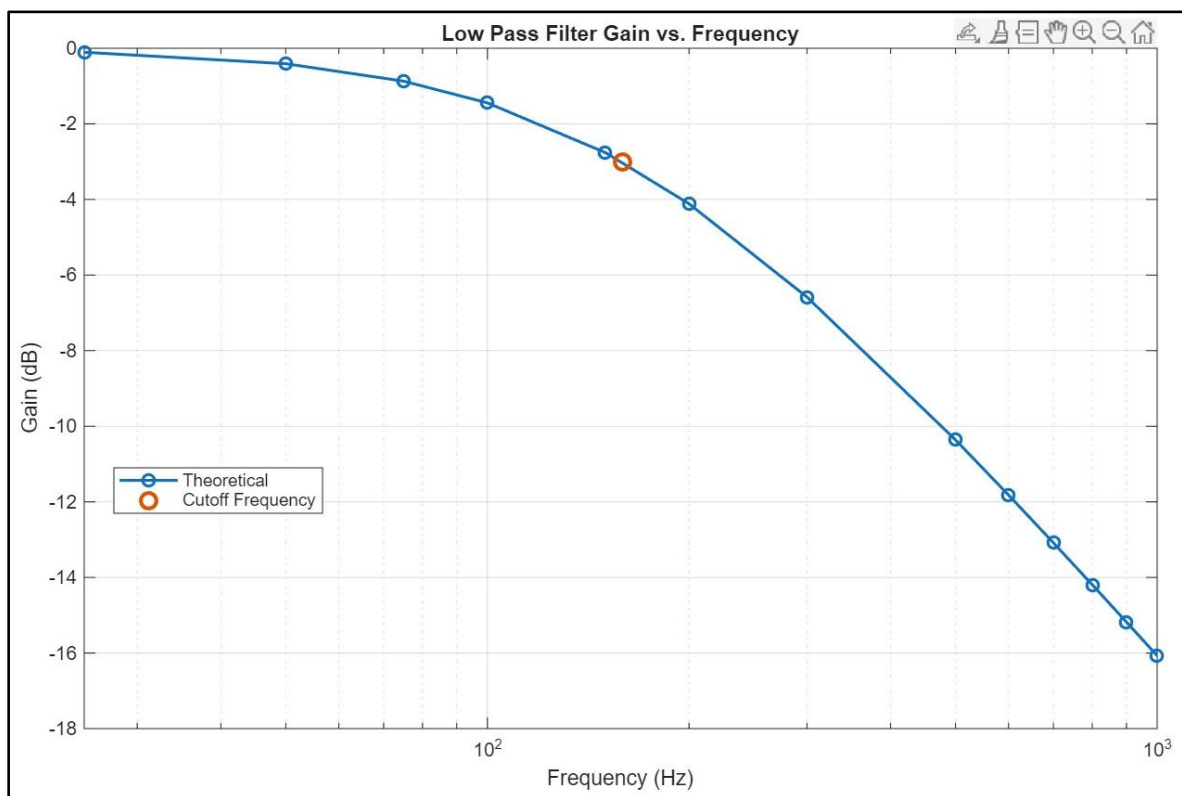


Figure #1: Gain vs. Frequency (LPF)

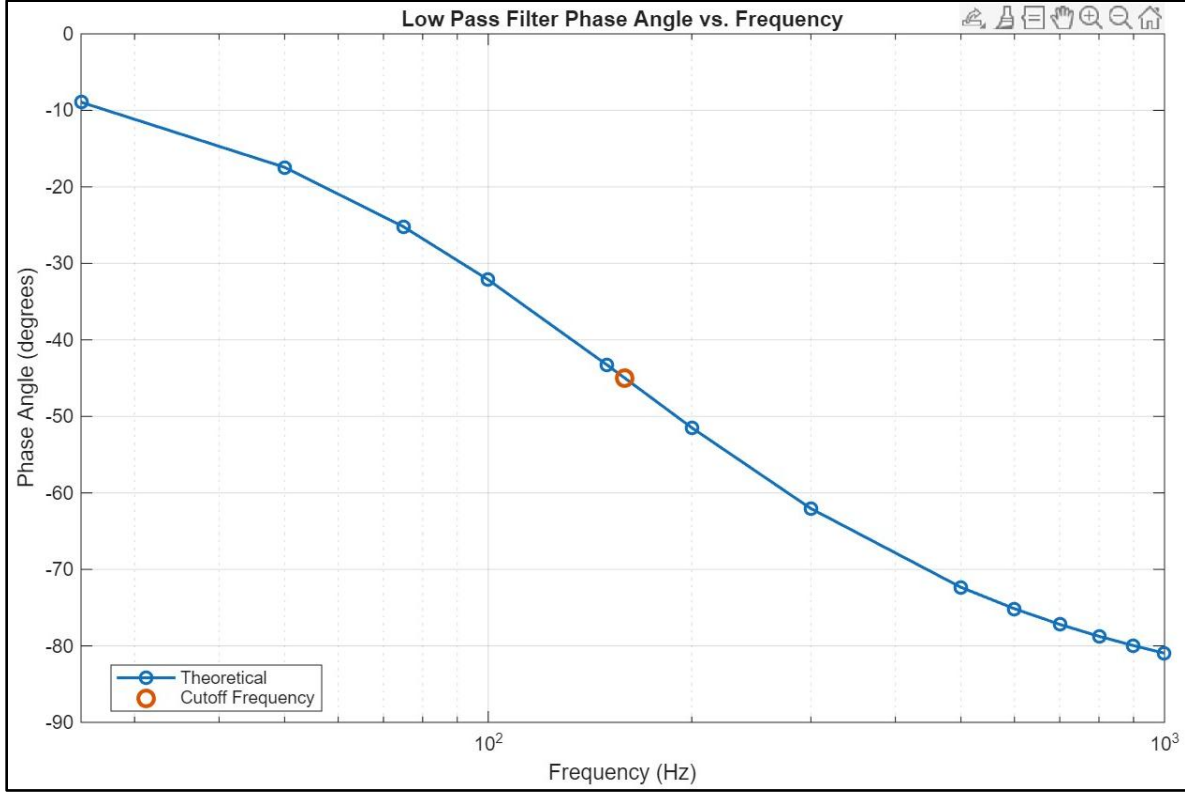


Figure #2: Phase Angle vs. Frequency (LPF)

Equations 7 and 8, given in the lab manual, were used to find the gain and phase angle for a high pass filter with $V_{in} = 4V$, $R = 1k\Omega$, and $C = 1\mu F$. Equation 9 was used to find the cutoff frequency. It is important to note that Equation 9 matches Equation 3, which was used to find the cutoff frequency of the low pass filter.

$$\left| \frac{V_{out}}{V_{in}} \right| = \frac{2\pi fRC}{\sqrt{1 + (2\pi fRC)^2}} \Rightarrow G = 20 \log \left(\frac{2\pi fRC}{\sqrt{1 + (2\pi fRC)^2}} \right) dB \quad (7)$$

$$\theta = \tan^{-1} \left(\frac{1}{2\pi fRC} \right) \quad (8)$$

$$f_c = \frac{1}{2\pi RC} \quad (9)$$

Equations 10 and 11 show equations 7 and 8 evaluated at $f = 25Hz$. These calculations were then repeated for varying values of f and used to complete Table 2. Equation 12 shows the cutoff frequency calculation. Figures 3 and 4 show the plots of gain and phase angle versus frequency.

$$G = 20 \log \left(\frac{2\pi * 25 * 1000 * 10^{-6}}{\sqrt{1 + (2\pi * 25 * 1000 * 10^{-6})^2}} \right) dB = -16.1835 \quad (10)$$

$$\theta = \tan^{-1} \left(\frac{1}{2\pi * 25 * 1000 * 10^{-6}} \right) = 81.0730 \quad (11)$$

$$f_c = \frac{1}{2\pi * 1000 * 10^{-6}} = 159.1594 \text{ Hz} \quad (12)$$

f (Hz)	Theoretical	
	Gain (dB)	Phase Angles (degrees)
25	-16.1835	81.0730
50	-10.4658	72.5594
75	-7.4061	64.7684
100	-5.4815	57.8581
150	-3.2752	46.6962
200	-2.1305	38.5119
300	-1.0770	27.9467
500	-0.4191	17.6568
600	-0.2953	14.8561
700	-0.2189	12.8092
800	-0.1686	11.2517
900	-0.1337	10.0284
1000	-0.1086	9.0431

Table #2: High Pass Filter Theoretical Results

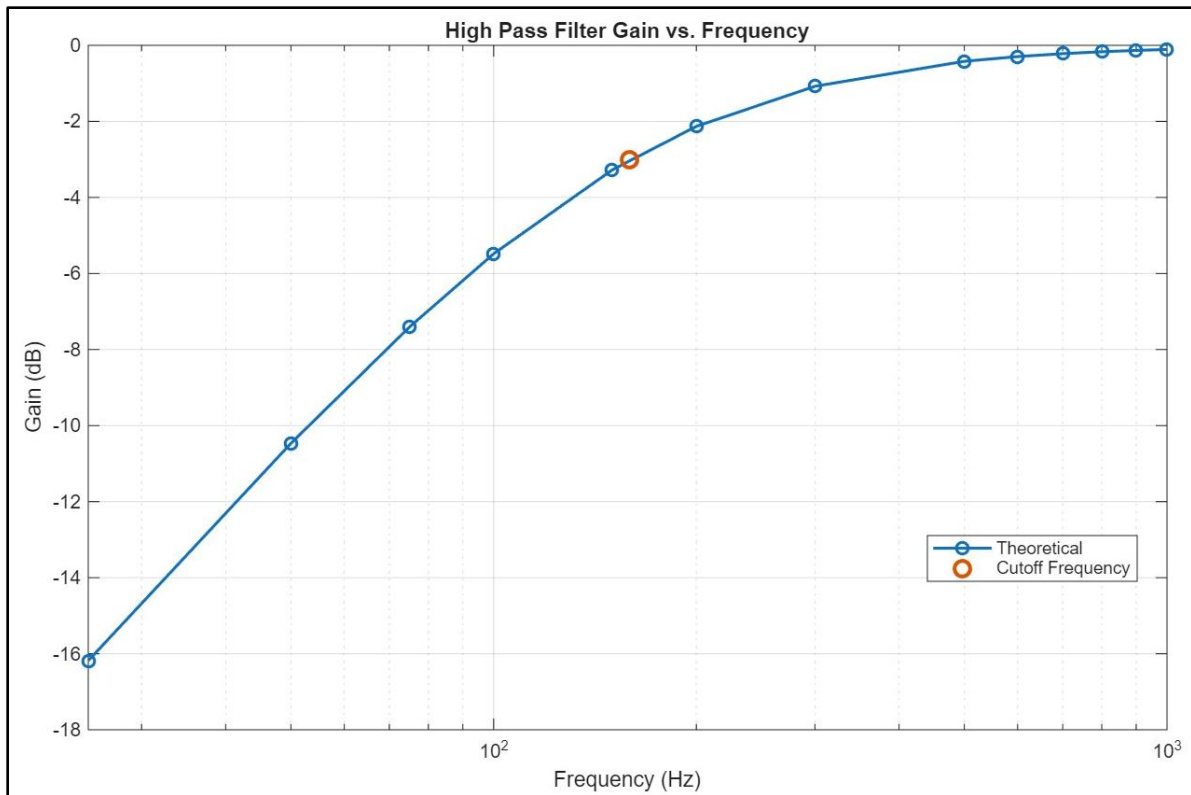


Figure #3: Gain vs. Frequency (HPF)

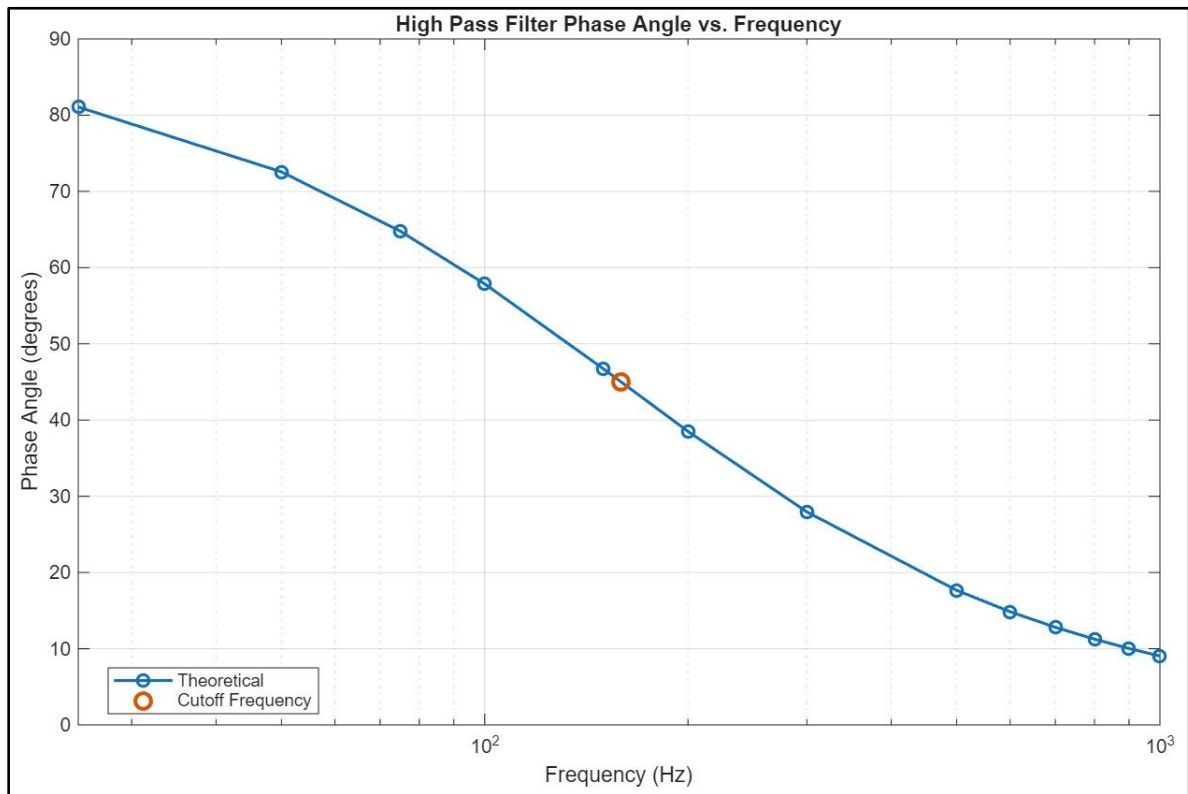


Figure #4: Phase Angle vs. Frequency (HPF)

Simulations and Results:

LTSpice was used to simulate both a low pass and high pass filter with $V_{in} = 4V_{pk-pk}$, $R = 1k\Omega$, and $C = 1\mu F$. Figures 5 and 6 show the schematics given in the lab manual for the LPF and HPF.

Figure 7 shows the LTSpice schematic used to simulate both filters simultaneously. Figure 8 shows the Bode plot for both simulated filters with the simulated -3dB point marked. Tables 3 and 4 show the simulation results for the LPF and HPF respectively.

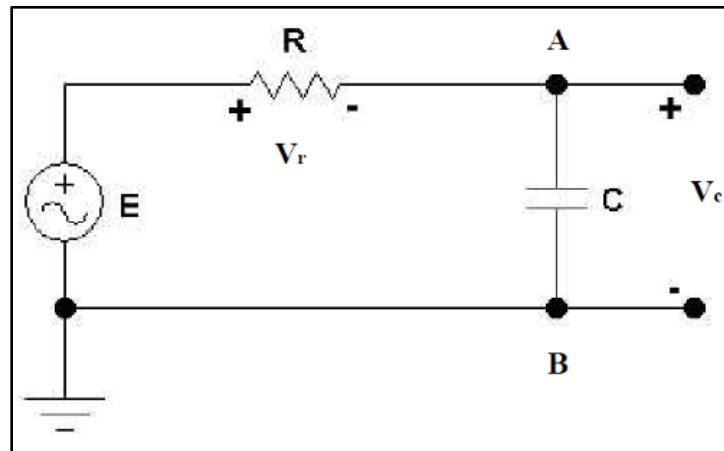


Figure #5: Low Pass Filter Schematic (Lab Manual)

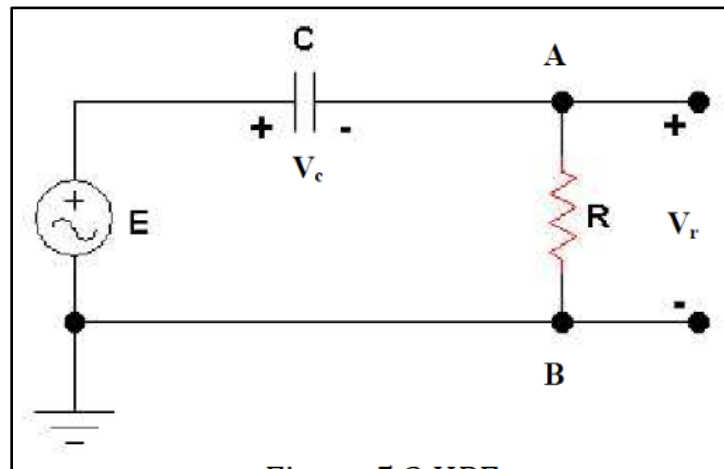


Figure #6: Low Pass Filter Schematic (Lab Manual)

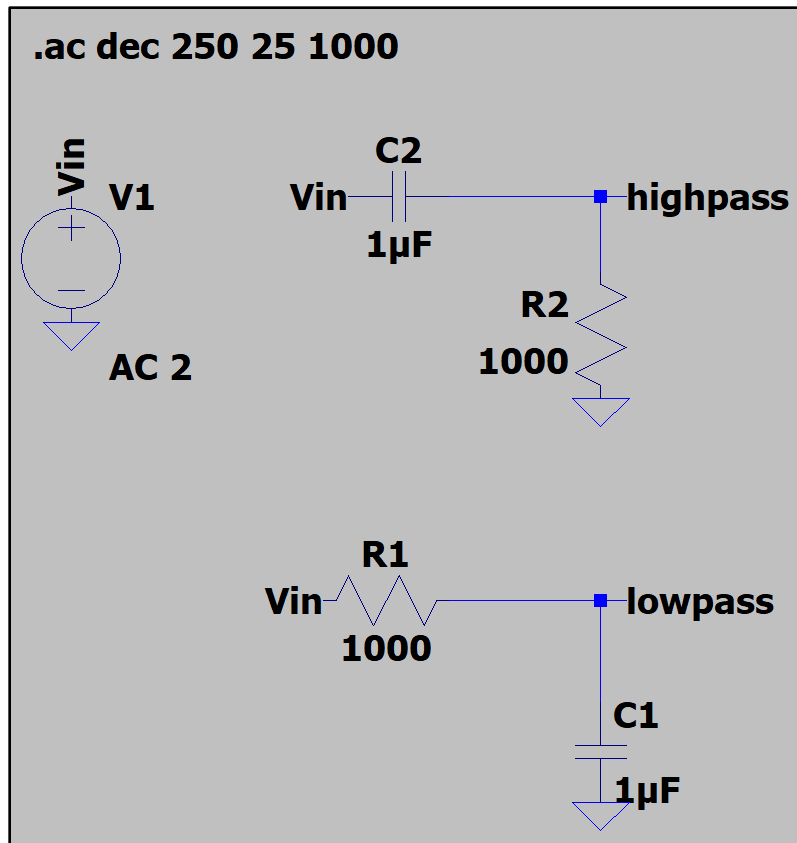


Figure #7: LTSpice Schematic (LPF and HPF)

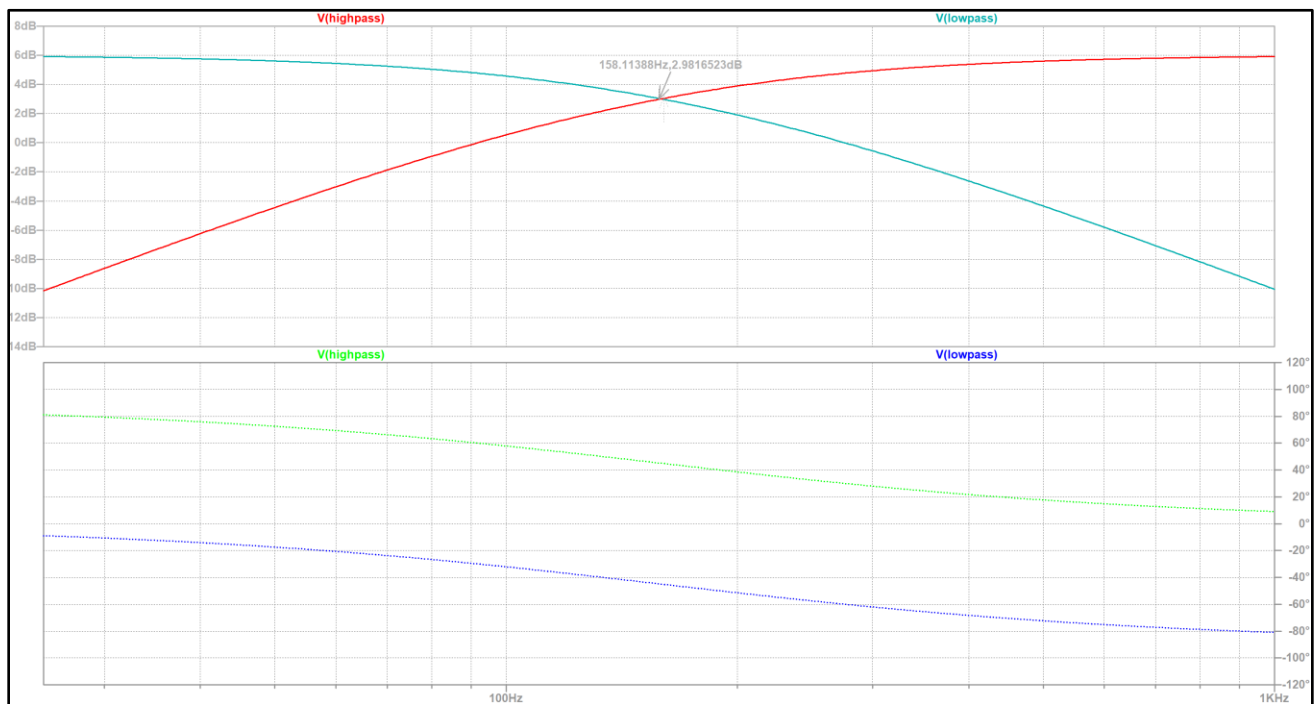


Figure #8: LTSpice Bode Plot (LPF and HPF)

	Simulation	
f (Hz)	Gain (dB)	Phase Angles (degrees)
25	-0.10586154	-8.9271434
50	-0.4087871	-17.440703
75	-0.8709666	-25.231744
100	-1.4450701	-32.141908
150	-2.7606321	-43.303809
200	-4.114777	-51.488062
300	-6.5830569	-62.053199
500	-10.362149	-72.343103
600	-11.82194	-75.143819
700	-13.084461	-77.190656
800	-14.193975	-78.748199
900	-15.182185	-79.971464
1000	-16.072235	-80.956939

Table #3: Low Pass Filter Simulation Results

	Simulation	
f (Hz)	Gain (dB)	Phase Angles (degrees)
25	-16.183464	81.072857
50	-10.46579	72.559297
75	-7.406144	64.768256
100	-5.4814727	57.858092
150	-3.2752096	46.696191
200	-2.1305798	38.511938
300	-1.0770344	27.946801
500	-0.41915173	17.656897
600	-0.29531781	14.856181
700	-0.21890279	12.809344
800	-0.16857801	11.251801
900	-0.13373748	10.028536
1000	-0.1086379	9.0430611

Table #4: High Pass Filter Simulation Results

Experimental Setup and Results:

The low pass and high pass filters in Figures 5 and 6 were constructed using a resistor, a capacitor, and a breadboard. The experimental setups for both LPF and HPF can be found in Figures 9 and 10 respectively. The Bode plots for both filters can be found in Figures 11 and 12. The experimental results for both filters can be found in Tables 5 and 6.

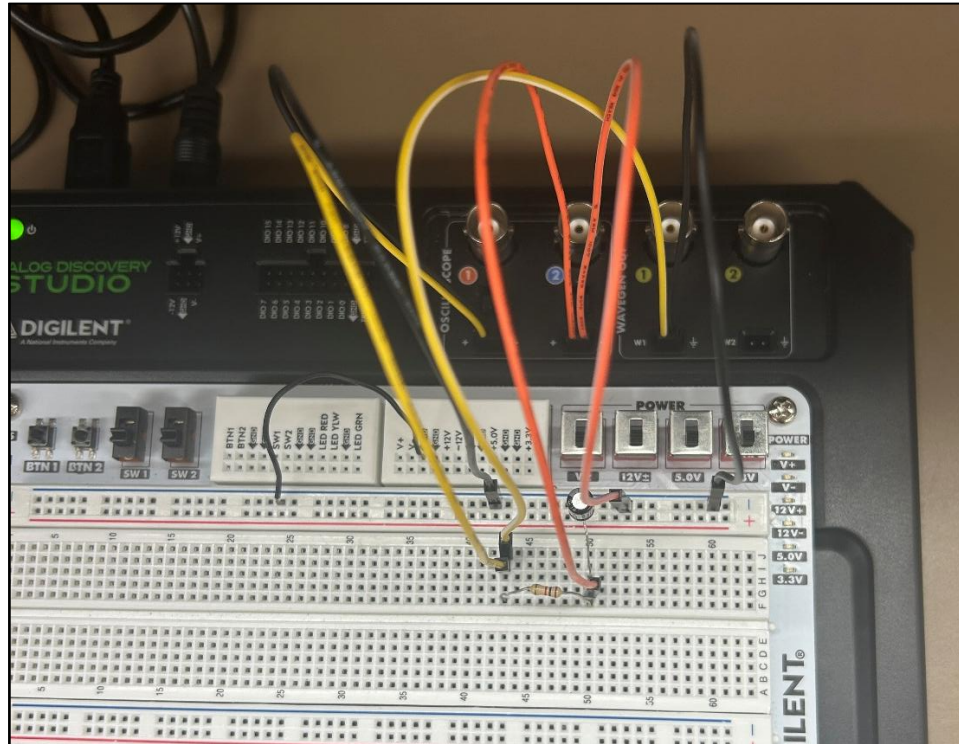


Figure #9: Low Pass Filter Experimental Setup

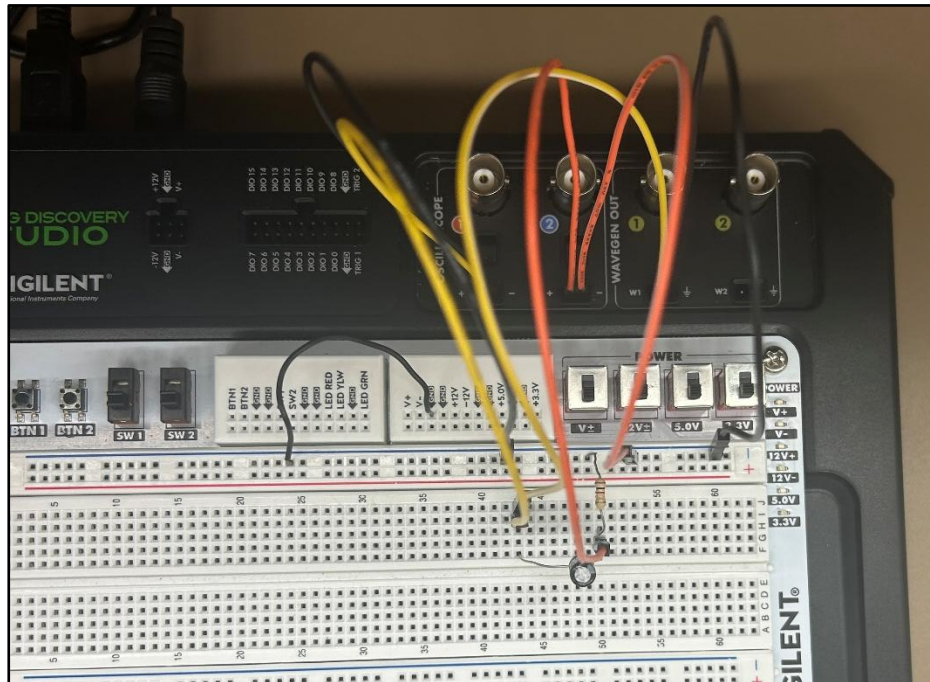


Figure #10: High Pass Filter Experimental Setup

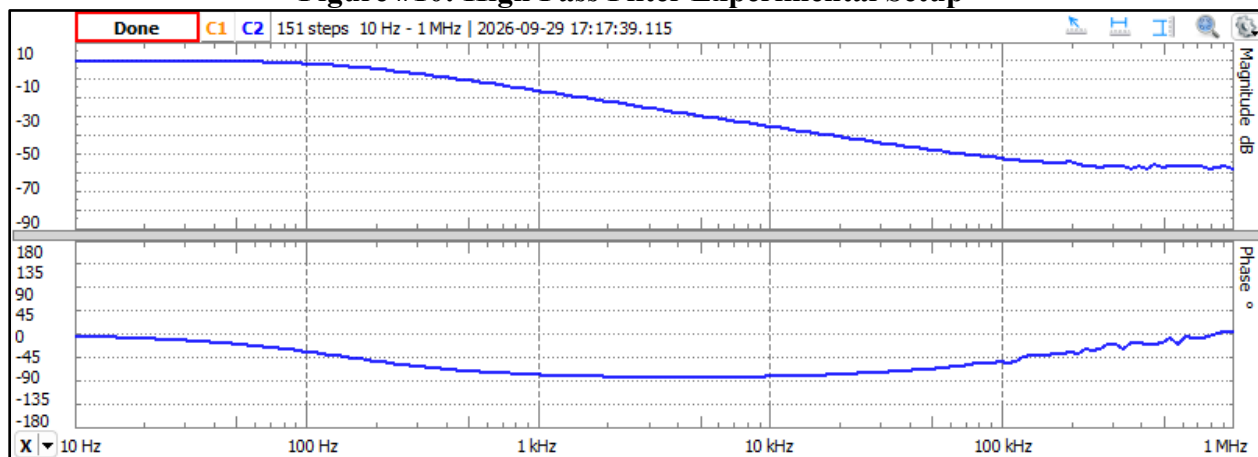


Figure #11: Low Pass Filter Bode Plot

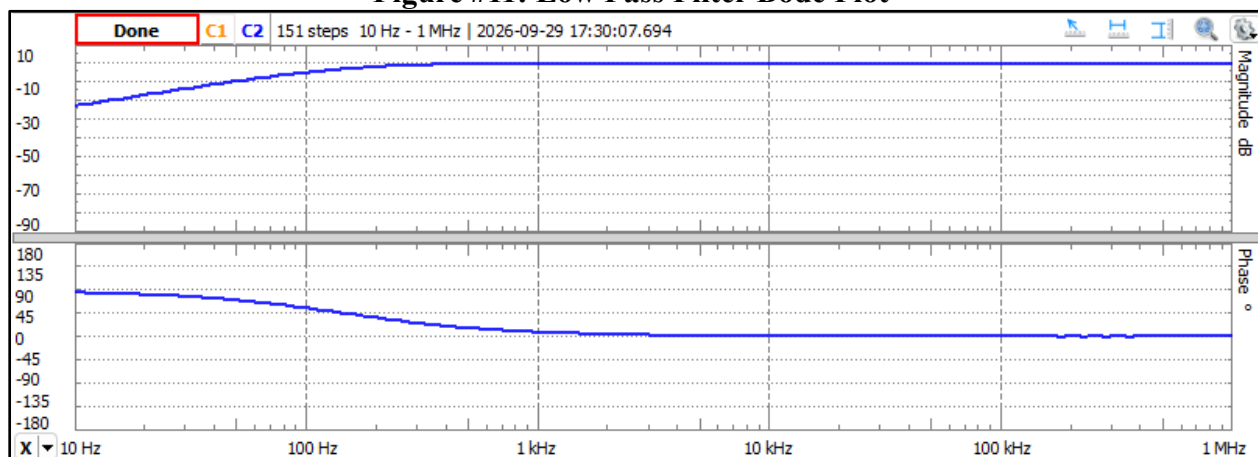


Figure #12: High Pass Filter Bode Plot

	Experimental	
f (Hz)	Gain (dB)	Phase Angles (degrees)
25	-0.1559	-9.65098
50	-0.5378	-18.54537
75	-1.0819	-26.46138
100	-1.7268	-33.28762
150	-3.1442	-43.9651
200	-4.541	-51.5398
300	-7.0072	-61.06341
500	-10.6832	-70.16734
600	-12.0914	-72.6128
700	-13.3067	-74.39027
800	-14.372	-75.74325
900	-15.3224	-76.78301
1000	-16.1778	-77.60788

Table #5: Low Pass Filter Experimental Results

	Experimental	
f (Hz)	Gain (dB)	Phase Angles (degrees)
25	-15.4241	78.864
50	-9.8709	69.852
75	-6.9518	61.797
100	-5.144	54.83
150	-3.1148	43.899
200	-2.0742	36.111
300	-1.1192	26.28
500	-0.5001	16.804
600	-0.3775	14.225
700	-0.2987	12.329
800	-0.2443	10.876
900	-0.2057	9.7364
1000	-0.177	8.8067

Table #6: High Pass Filter Experimental Results

Comparison and Discussion:

The theoretical and simulation results are nearly identical, the only difference being the number of decimal places the results were rounded to. The experimental results are also remarkably similar to the simulation values with only slight deviations. These differences are likely due to non-ideal components and tolerances in the resistor and capacitor. A summary of the results and percent error (experimental relative to simulation) for both filters can be found in Tables 7 and 8.

f (Hz)	Simulation		Experimental		Percent Error	
	Gain (dB)	Phase Angles (degrees)	Gain (dB)	Phase Angles (degrees)	Gain (dB)	Phase Angles (degrees)
25	-0.1058615	-8.9271434	-0.1559	-9.65098	47.27%	8.11%
50	-0.4087871	-17.440703	-0.5378	-18.54537	31.56%	6.33%
75	-0.8709666	-25.231744	-1.0819	-26.46138	24.22%	4.87%
100	-1.4450701	-32.141908	-1.7268	-33.28762	19.50%	3.56%
150	-2.7606321	-43.303809	-3.1442	-43.9651	13.89%	1.53%
200	-4.114777	-51.488062	-4.541	-51.5398	10.36%	0.10%
300	-6.5830569	-62.053199	-7.0072	-61.06341	6.44%	1.60%
500	-10.362149	-72.343103	-10.6832	-70.16734	3.10%	3.01%
600	-11.82194	-75.143819	-12.0914	-72.6128	2.28%	3.37%
700	-13.084461	-77.190656	-13.3067	-74.39027	1.70%	3.63%
800	-14.193975	-78.748199	-14.372	-75.74325	1.25%	3.82%
900	-15.182185	-79.971464	-15.3224	-76.78301	0.92%	3.99%
1000	-16.072235	-80.956939	-16.1778	-77.60788	0.66%	4.14%

Table #7: Low Pass Filter Results and Percent Error

f (Hz)	Simulation		Experimental		Percent Error	
	Gain (dB)	Phase Angles (degrees)	Gain (dB)	Phase Angles (degrees)	Gain (dB)	Phase Angles (degrees)
25	-16.183464	81.072857	-15.4241	78.864	4.69%	2.72%
50	-10.46579	72.559297	-9.8709	69.852	5.68%	3.73%
75	-7.406144	64.768256	-6.9518	61.797	6.13%	4.59%
100	-5.4814727	57.858092	-5.144	54.83	6.16%	5.23%
150	-3.2752096	46.696191	-3.1148	43.899	4.90%	5.99%
200	-2.1305798	38.511938	-2.0742	36.111	2.65%	6.23%
300	-1.0770344	27.946801	-1.1192	26.28	3.91%	5.96%
500	-0.41915173	17.656897	-0.5001	16.804	19.31%	4.83%
600	-0.29531781	14.856181	-0.3775	14.225	27.83%	4.25%
700	-0.21890279	12.809344	-0.2987	12.329	36.45%	3.75%
800	-0.16857801	11.251801	-0.2443	10.876	44.92%	3.34%
900	-0.13373748	10.028536	-0.2057	9.7364	53.81%	2.91%
1000	-0.1086379	9.0430611	-0.177	8.8067	62.93%	2.61%

Table #8: High Pass Filter Results and Percent Error

Conclusion:

This laboratory demonstrated how the frequency of an input signal affects the gain and phase angle of low-pass and high-pass filters. The theoretical calculations, simulations, and experimental measurements showed the expected filter behavior, with the gain changing as the

input frequency moved through the filter's cutoff region. The experimental results were generally close to the simulated values, with some differences due to component tolerances, measurement uncertainty, and limitations of the physical equipment. Overall, the results confirmed the frequency-dependent characteristics of the filters and demonstrated the relationship between cutoff frequency, gain, and phase angle.